Integrating SBAS-InSAR, Optical Imagery, and Machine Learning for Landslide Hazard Assessment: A *Case Study of Lagam Escarpment*

by

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Department of Geomatic Engineering and Geospatial Information Systems (GEGIS)

DECLARATION

I declare that this project is my own work and has not been submitted by anybody else in any other university for the award of any degree to the best of my knowledge.

Sign…… Date……..

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CERTIFICATION

This project has been submitted for examination with my approval as the candidate's supervisor.

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ABSTRACT

Landslides pose significant threats to human life, infrastructure, and the environment, particularly in regions with steep terrains and heavy rainfall, such as the Lagam Escarpment in Kenya. Traditional landslide detection methods are often limited by accessibility, cost, and temporal coverage. This study leverages advanced remote sensing techniques and machine learning to enhance landslide susceptibility mapping in the Lagam Escarpment.

Sentinel-1 Synthetic Aperture Radar (SAR) data was processed using Small Baseline Subset Interferometric SAR (SBAS-InSAR) to monitor ground displacement, while Sentinel-2 optical imagery was employed to detect surface changes through spectral indices such as NDVI, NDWI, BSI, and BR. The integration of these datasets enabled the creation of a comprehensive landslide inventory. Machine learning models; Random Forest (RF) and Support Vector Machine (SVM), were then applied to generate landslide susceptibility maps, which classified the study area into five risk levels. The RF model achieved an Area Under the Curve (AUC) of 0.97, outperforming SVM (AUC = 0.95), demonstrating high predictive accuracy. Key findings revealed deformation rates of -4 to 3 cm/year, with severe displacement along the escarpment.

The study highlights the effectiveness of combining SBAS-InSAR, optical remote sensing, and machine learning for landslide hazard assessment. By integrating various geospatial datasets, including topographic, climatic, geological, and anthropogenic factors, the study enhances predictive modeling for landslide-prone areas. The findings provide valuable insights for disaster risk reduction, urban planning, infrastructure development, and environmental management, demonstrating the importance of remote sensing and machine learning techniques in landslide prediction and mitigation strategies.

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Acronyms and abbreviations

1. **ALOS PALSAR** – Advanced Land Observing Satellite Phased Array type L-band Synthetic Aperture Radar
2. **ASF -** Alaska Satellite Facility
3. **AUC -** Area Under the Curve
4. **BR** – Brightness Ratio
5. **BSI** – Bare Soil Index
6. **CHIRPS -** Climate Hazards Group InfraRed Precipitation with Station data
7. **CNN** – Convolutional Neural Network
8. **DEM** – Digital Elevation Model
9. **FPR -** False Positive Rate
10. **GEE** – Google Earth Engine
11. **GLCM -** Gray Level Co-occurrence Matrix
12. **InSAR** – Interferometric Synthetic Aperture Radar
13. **LULC -** Land Use Land Cover
14. **NDVI** – Normalized Difference Vegetation Index
15. **NDWI** – Normalized Difference Water Index
16. **RF** – Random Forest
17. **RNN** – Recurrent Neural Network
18. **ROC** – Receiver Operating Characteristic
19. **SAR** – Synthetic Aperture Radar
20. **SBAS** – Small Baseline Subset
21. **SLC -** Single Look Complex
22. **SMOTE** – Synthetic Minority Over-sampling Technique
23. **SPI** – Stream Power Index
24. **SRTM -** Shuttle Radar Topography Mission
25. **SVM** – Support Vector Machine
26. **TPR -** True Positive Rate
27. **TWI -** Topographic Wetness Index
28. **USGS-** United States Geological Survey.

1. INTRODUCTION

1.1 Background

Landslides are defined as the movement of a mass of rock, debris or earth down a slope, and can result in enormous casualties and huge economic losses (Zhao & Lu, 2018). They are often triggered by factors such as intense rainfall, earthquakes, volcanic activity, or human-induced alterations like deforestation and construction. These events pose significant threats to human life, infrastructure, and the environment, leading to casualties, economic losses, and land degradation (Perera et al., 2018).

Accurate quantification and mapping of landslides are essential for effective risk assessment and mitigation. The Unified Global Landslide Database (UGLD), with 161 countries, has 37,946 landslides, and 185,753 fatalities registered between 1903 and 2020 (Gómez et al., 2023). Due to the lack of existing records, landslides in many developing countries remain undocumented, leading to their exclusion from such global analyses, and for those who attempt, they rely on traditional methods.

Traditional methods have relied on field surveys and aerial photography, which, while valuable, are often constrained by accessibility challenges, high costs, and limited temporal coverage. The advent of remote sensing technologies has revolutionized landslide detection, offering broader spatial coverage and more frequent data acquisition (Novellino et al., 2024).

Optical remote sensing utilizes sensors that capture reflected sunlight to produce high-resolution images of the Earth's surface. These images are instrumental in identifying surface features indicative of landslides, such as scarps, displaced material, vegetation changes and other satellite-derived land cover information(Satriano et al., 2023). However, optical sensors are hindered by atmospheric conditions like cloud cover and require daylight, limiting their effectiveness in continuous monitoring (Mondini et al., 2019).

In contrast, radar sensors, particularly Synthetic Aperture Radar (SAR), emit microwave signals to penetrate cloud cover and operate independently of lighting conditions. This capability enables the detection of ground surface changes associated with landslides, such as soil displacement and deformation, even in adverse weather or during nighttime (Mondini et al., 2021).

Integrating optical and radar data enhances landslide detection and mapping by leveraging the strengths of both sensor types. For instance, combining optical imagery's detailed surface characterization with radar's all-weather, day-and-night capabilities provides a more comprehensive analysis of landslide-prone areas (Ghorbanzadeh et al., 2020). Studies have demonstrated that such integration improves the accuracy of landslide susceptibility models, offering valuable insights for disaster management and mitigation strategies (Ranasinghe et al., 2019).

The integration of machine learning (ML) techniques with geospatial data has also shown great promise in landslide susceptibility mapping. Machine learning algorithms, such as Random Forest and Support Vector Machines, can process large datasets, identify complex patterns, and improve prediction accuracy (Pham et al., 2021). For instance, (Lei et al., 2020)demonstrated the effectiveness of integrating artificial intelligence and statistical modeling for spatial prediction. These studies highlight the importance of combining multiple data sources and advanced analytical techniques for enhanced landslide hazard assessment.

In summary, the synergy between optical and radar remote sensing technologies has significantly advanced the field of landslide detection and susceptibility mapping (Q. Xu et al., 2023). By combining diverse data sources and analytical methods, researchers can develop more accurate and reliable models, contributing to enhanced preparedness and resilience against landslide hazards.

1.2 Motivation and Problem Statement

Traditional methods for landslide detection and susceptibility mapping often rely on ground surveys, historical records, and empirical models, which have several limitations. These approaches are time-consuming, expensive, and constrained by accessibility challenges in remote or hazardous terrains (Kim et al., 2024). Furthermore, in many developing countries, the absence of systematic landslide inventories limits the accuracy of susceptibility assessments (Aristizábal-Giraldo & Ruiz-Vásquez, 2023).Given the dynamic nature of landslides, traditional approaches also struggle to capture real-time changes, making them inadequate for early warning systems and disaster response. Consequently, a more efficient and scalable approach is required to enhance landslide detection and risk assessment.

Optical and radar remote sensing techniques offer a promising alternative by providing continuous, high-resolution, and multi-temporal observations for landslide monitoring. Optical sensors such as Sentinel-2 allow for spectral analysis of surface changes using several indices, while radar-based techniques, particularly Sentinel 1, enable the detection of ground deformation even in cloud-covered or vegetated regions. This synergy between optical and radar data improves the accuracy of landslide susceptibility maps and ensures the identification of both active and potential landslide zones (Chrysafi et al., 2024). Moreover, SBAS InSAR has demonstrated its effectiveness in monitoring slow-moving landslides by detecting millimeter-scale ground displacement, an advantage over traditional methods that often fail to capture such subtle changes (S. Li et al., 2022).

In Elgeyo Marakwet, landslides have been recurrent, with significant socio-economic and environmental impacts (Kiama, n.d.). Historical data indicate severe landslide events, including the 2019 Kaben landslide that resulted in multiple fatalities and the displacement of hundreds

of residents (Jepkemboi Loice et al., 2021). These events underscore the urgent need for advanced monitoring techniques to mitigate future risks. The approach adopted in this study addresses the limitations of existing methods by providing a cost-effective since open source and free data or tools are used, and scalable way for landslide susceptibility mapping. The results of this research will contribute to improved disaster preparedness, early warning systems, and sustainable land-use planning in Elgeyo Marakwet and other landslide-prone regions.



The house in Elgeyo-Marakwet where a family of four died Thursday night after a landslide. PHOTO | JARED NYATAYA | NATION MEDIA GROUP

Figure 1: Landslide in Lagam escarpment, Elgeyo-Marakwet that occurred on October 18, 2019

1.3 Research Identification and Objectives

The main objective of this research is to develop a comprehensive landslide susceptibility assessment by integrating Sentinel-1 and Sentinel-2 imagery, along with machine learning techniques, to improve landslide detection and prediction in the Lagam Escarpment. This is achievable through the following specific objectives:

1.3.1 Research Objectives

1. To apply SBAS-InSAR techniques on Sentinel-1 data for detecting and monitoring ground displacement in the Lagam Escarpment by obtaining the displacement maps and velocity map.
2. To map past landslides using a combination of SBAS-InSAR and optical imagery to establish a historical record of landslide occurrences.
3. To develop a landslide susceptibility map by integrating results from SBAS-InSAR and optical imagery landslide occurrences, and machine learning to assess landslide hazard in the study area.

1.3.2 Research Questions

The study aims to answer the following questions:

1. How effectively can SBAS-InSAR techniques applied to Sentinel-1 data detect and monitor ground displacement in the Lagam Escarpment, and what are the observed displacement trends over time?
2. To what extent can the combination of SBAS-InSAR and optical imagery accurately identify and map past landslide occurrences in the study area?
3. How can the integration of SBAS-InSAR results, optical imagery-based landslide mapping, and machine learning models improve the accuracy of landslide susceptibility assessments in the Lagam Escarpment?

1.4 Study Outline

This research is structured into five chapters. Chapter 1 introduces the study by outlining the background, motivation, problem statement, justification, research objectives, and research questions. Chapter 2 presents a comprehensive literature review, examining previous studies, theoretical frameworks, and methodological approaches related to landslide hazard assessment. Chapter 3 details the data sources and methodologies used, including SBAS-InSAR processing, optical imagery analysis, and machine learning techniques. Chapter 4 presents the results, such as displacement maps, landslide susceptibility maps, and key findings. Finally, Chapter 5 concludes the study by summarizing its key contributions and providing recommendations for future research in landslide hazard assessment.

2. LITERATURE REVIEW

2.1. Introduction to Landslide Hazard Assessment

Landslides are the downslope movement of rock, soil, and debris under the influence of gravity. They are geomorphological processes that shape the landscapes of all continents, dismantling mountains and contributing sediments to the river networks (Mondini et al., 2021). They are among the most destructive natural hazards, often triggered by a combination of geological, hydrological, and anthropogenic factors. The speed at which they occur can vary considerably, from just a few cm per year up to tens of km/h and the volume of the displaced material can be anywhere from a few tens of m^3 to several km^3 (Haque et al., 2019).

Landslide hazard assessment is the process of analyzing and evaluating the causes and likelihood of landslides and their possible consequences (Shan et al., 2024). Landslide Susceptibility Mapping involves a comprehensive analysis of various factors, including geological environmental factors, historical landslides and the landslide physical laws of an area, to determine the probability of future landslide events in that area (Reichenbach et al., 2018).

2.1.1 Causes of Landslides

Landslides are caused by natural and human-induced factors, including intense or prolonged rainfall, seismic shaking, volcanic activity, and rapid snow melting. The occurrence of landslides is affected by several factors such as topography, geology, hydrological, land cover and human activity (F. Huang et al., 2024).

Natural causes include water saturation from rainfall, snowmelt, and groundwater fluctuations, which increase pore water pressure, reducing the shear strength of slopes and triggering mass movements, leading to landslides, debris flows, and floods. Weak or unstable structures, such as fault zones, fractured rocks, or clay-rich soils are prone to failure when saturated by water. Seismic activity can also induce landslides by shaking loose soil, causing liquefaction, or triggering slope failures, as seen in the 1964 Alaska earthquake. Volcanic activity further contributes to landslides through lahars; rapid flows of volcanic debris mixed with water. Human activities, such as deforestation, poor drainage management, construction on unstable land, and excavation, can significantly increase landslide risk by destabilizing slopes and altering natural drainage patterns. Proper engineering and planning can mitigate these risks, preventing severe damage and loss of life (Jaboyedoff et al., 2019), Highland et al., 2008).

Landslides impact both the built and natural environments, often intersecting in areas like agricultural and logged forest lands. In the built environment, landslides can cause severe damage to residential and commercial structures, utilities, and transportation networks.

Fast-moving landslides, such as debris flows, are particularly destructive, while slow-moving ones may allow time for mitigation but can still lead to structural failure over time. Landslides also disrupt lifelines such as roads and railways, posing risks to commerce, tourism, and emergency response. In the natural environment, landslides reshape landscapes, affecting forests, grasslands, and wildlife habitats. While they often cause temporary ecological damage, some landslides can create new habitats or even benefit ecosystems in the long term. Specific types of natural landslides include submarine landslides, which can trigger tsunamis; coastal cliff erosion, which threatens marine life and human settlements; and landslide dams, which can create temporary lakes but also pose flood risks if they fail. As populations expand into landslide-prone areas, the risks increase, highlighting the need for improved land-use policies and mitigation strategies (Highland et al., 2008).

2.1.2 Importance of Landslide Hazard Assessment

Landslide hazard assessment is critical for disaster risk reduction, as landslides pose significant threats to human lives, infrastructure, agriculture, and ecosystems (Alcántara-Ayala & Sassa, 2023). Accurate identification of landslide-prone areas enables the implementation of mitigation measures, such as slope stabilization, early warning systems, and land-use planning, which can save lives and reduce economic losses. In rapidly urbanizing regions, landslide hazard assessment is particularly important for guiding sustainable development and ensuring that infrastructure projects avoid high-risk areas (Kiama, n.d.). Furthermore, as climate change continues to alter precipitation patterns and increase the frequency of extreme weather events, landslide hazard assessment becomes an essential tool for climate change adaptation Gariano & Guzzetti, 2016). By providing actionable insights into landslide risks, this study contributes to safeguarding vulnerable communities and promoting resilient development in landslide-prone regions, with landslide inventory mapping and landslide susceptibility mapping serving as the essential first steps in this process.

2.1.3 Overview of Different Approaches to Landslide Mapping

Landslide mapping is categorized into two key approaches: landslide inventory mapping and landslide susceptibility mapping.

Landslide inventory mapping focuses on identifying and documenting past landslide occurrences, providing crucial baseline data for further analysis. Traditionally, landslide inventories have been created through field surveys and the manual interpretation of aerial photographs. These methods, while valuable, are time-consuming, labor-intensive, and often limited by accessibility challenges in remote or hazardous terrains. Additionally, field surveys may be subjective, leading to inconsistencies in landslide identification and classification. The reliance on historical records and eyewitness reports further introduces gaps and potential inaccuracies in the dataset (Hussain et al., 2022).

To overcome these limitations, modern remote sensing techniques have been increasingly adopted for landslide inventory mapping. Optical satellite imagery enables the detection of surface changes, while Interferometric Synthetic Aperture Radar (InSAR) is particularly useful for monitoring ground deformation over time, even in cloud-covered or vegetated areas. LiDAR technology further enhances landslide mapping by providing high-resolution digital elevation models that reveal subtle topographic features indicative of past slope failures (Aslan et al., 2020). Automated classification techniques using machine learning and GIS-based spatial analysis further improve the efficiency and accuracy of inventory mapping. By integrating these remote sensing approaches, landslide inventories become more comprehensive and reliable, serving as a strong foundation for susceptibility assessments and risk mitigation strategies (F. Huang et al., 2024).

Landslide susceptibility mapping, on the other hand, predicts areas prone to future landslides based on environmental and triggering factors (Shan et al., 2023). Various methods are used for this purpose. Traditional statistical models, such as logistic regression and probabilistic approaches, establish relationships between past landslides and causative factors like slope, soil type, and rainfall (Conforti & Ietto, 2021). Deterministic models rely on the physics of slope failure, employing techniques like finite element modeling and limit equilibrium analysis for precise but data-intensive assessments. Machine learning models, including Support Vector Machines (SVM), Random Forest (RF), and Artificial Neural Networks (ANNs), leverage large datasets encompassing terrain attributes, land cover, and climatic data for more data-driven predictions (Liu et al., 2023). Remote sensing-based approaches integrate satellite imagery and radar data for continuous monitoring of landslide-prone areas. Finally, hybrid approaches combine multiple techniques such as integrating SBAS-InSAR velocity maps with optical imagery or machine learning algorithms with remote sensing data to enhance accuracy and scalability. These comprehensive mapping strategies are essential for effective landslide hazard assessment and mitigation.

2.2. Remote Sensing Techniques for Landslide Detection

Remote sensing has become an essential tool for landslide detection, offering efficient and scalable methods for mapping and monitoring slope failures over large areas. Unlike traditional field-based approaches, remote sensing provides consistent and repeatable observations, enabling the detection of both past and ongoing landslides with improved accuracy (Hussain et al., 2022). By utilizing various sensors, including optical, radar, and LiDAR, remote sensing techniques can capture surface changes, ground deformation, and topographic variations associated with landslides (Zhong et al., 2020). The methods used in this study are discussed below.

2.2.1 Optical Remote Sensing

Optical remote sensing involves the use of satellite imagery that captures visible and near-infrared radiation reflected from the Earth's surface. This technique is widely used in landslide detection and mapping due to its ability to provide high-resolution spatial data on land cover,

vegetation, and surface features (Lu et al., 2024). The most commonly used optical sensors for landslide monitoring include Sentinel-2 and Landsat, which offer multispectral imagery with resolutions ranging from 10 to 30 meters (Amatya et al., 2019).

Optical imagery is particularly effective for detecting changes in land cover and surface morphology, which are critical for identifying landslide-prone areas. By comparing pre- and post-event images, shifts in vegetation, soil composition, and landscape morphology can be observed to detect landslides, (Ghorbanzadeh et al., 2020). Sentinel-2, with its high spatial resolution (10 m) and frequent revisit time (5 days), is especially suitable for monitoring small-scale landslides and changes in surface conditions (P. Lu et al., 2019, Amatya et al., 2019). Landsat imagery, although coarser in resolution (30 m), provides a long-term historical record valuable for analyzing trends in landslide activity over time (Ghorbanzadeh et al., 2020, Li et al., 2024).

Optical imagery is useful for detecting changes in land cover and surface morphology, which are critical for identifying landslide-prone areas. For example, pre- and post-event images can be compared to detect landslides through shifts in vegetation, soil type, and landscape morphology (Kyriou et al, 2020). Sentinel-2, with its high spatial resolution (10 m) and frequent revisit time (5 days), is especially effective for monitoring small-scale landslides and changes in surface conditions (Ghorbanzadeh et al., 2020). Landsat imagery, although coarser in resolution (30 m), provides a long-term historical record that is valuable for analyzing trends in landslide activity over time (Chen et al., 2019). To enhance landslide detection, spectral indices such as the Normalized Difference Vegetation Index (NDVI), Bare Soil Index (BSI), and Brightness Ratio are often compared before and after events to delineate affected areas (Lu et al., 2024,Ghorbanzadeh et al., 2020).

The primary advantage of optical remote sensing is its ability to provide detailed information on surface features, such as slope gradients, drainage networks, and vegetation cover, which are critical for landslide susceptibility mapping (Amatya et al., 2019). However, optical imagery is limited by its susceptibility to cloud cover, which can obscure the Earth’s surface, especially in tropical and subtropical regions (Chen et al., 2019). Additionally, optical sensors are less effective in areas with dense vegetation or complex terrain, where surface changes may not be clearly visible (Ghorbanzadeh et al., 2020).

2.2.2 Radar Remote Sensing

Radar remote sensing, particularly Synthetic Aperture Radar (SAR), has emerged as a powerful tool for detecting and monitoring landslides, especially in regions with frequent cloud cover or complex topography. SAR sensors, such as those on Sentinel-1, emit microwave signals that

penetrate clouds and provide detailed information on surface deformation, making them ideal for landslide detection (Ghorbanzadeh et al., 2020).

SAR works by transmitting microwave signals toward the Earth's surface and recording the reflected signals to generate high-resolution images of the terrain. Interferometric SAR (InSAR) is an advanced technique that measures the phase difference between two SAR images taken at different times, enabling the detection of ground deformation at millimeter-scale precision (Shan et al., 2024). This makes InSAR particularly effective for monitoring slow-moving landslides, which are often precursors to larger slope failures (Dai et al., 2022) .

The Small Baseline Subset (SBAS) technique is a variation of InSAR that uses a series of SAR images over time to monitor surface displacements with high temporal and spatial resolution (Xu et al., 2024). SBAS-InSAR is particularly useful for detecting slow-moving ground deformation, which is critical for monitoring landslides in areas like the Lagam Escarpment, where landslides may develop gradually over time (Chen et al., 2024). Unlike optical imagery, SBAS-InSAR is not affected by cloud cover or daylight conditions, making it a reliable tool for continuous monitoring of landslide activity (Ghorbanzadeh et al., 2020).

Comparison with Other InSAR Techniques

While Differential InSAR (DInSAR) is effective for detecting large displacements over short periods, it is limited by the need for high-quality data and short temporal baselines (Li et al., 2022) . Persistent Scatterer Interferometry (PSI), on the other hand, is effective in urban areas but less suitable for natural terrains with sparse stable reflectors (Aslan et al., 2020). SBAS-InSAR overcomes these limitations by combining the strengths of both techniques, making it ideal for monitoring landslides in diverse environments (Chen et al., 2024).

2.2.3 Integration Between Optical and Radar Data

The integration of optical and radar remote sensing offers a more comprehensive approach to landslide detection and monitoring. While optical imagery provides detailed information on surface features and land cover changes, radar imagery captures ground deformation with high precision, even in challenging weather conditions (Ghorbanzadeh et al., 2020). This synergy allows for a more holistic understanding of landslide dynamics, from surface changes to subsurface movements.

Combining optical and radar imagery for landslide mapping offers several benefits. Enhanced accuracy is achieved by leveraging the strengths of both data types. Optical imagery identifies surface changes such as vegetation loss and land cover alterations, while radar imagery detects ground deformation, providing complementary insights (Ghorbanzadeh et al., 2020). Improved temporal coverage is another advantage, as radar imagery can penetrate cloud cover and

operate in all weather conditions, filling observational gaps left by optical sensors with limited revisit times (Kyriou et al., 2020).

A study by Liu et al. (2020) integrated Sentinel-1 and Sentinel-2 imagery to monitor landslide activity in the Himalayan region. The fusion of both data sources provided a more accurate identification of landslide boundaries and movement patterns than either data source alone. Similarly, a study in the Niulan River Basin combined SBAS-InSAR with optical imagery to track slow-moving landslides and map their impact on land cover, demonstrating the effectiveness of integrated approaches for landslide hazard assessment (Yang et al., 2023).

2.3. Machine Learning in Landslide Susceptibility Mapping

Machine learning (ML) has become a powerful tool for landslide susceptibility mapping due to its ability to analyze complex, non-linear relationships between landslide causative factors and environmental variables (Liu et al., 2023). Two of the most commonly used algorithms in landslide prediction are Support Vector Machines (SVM) and Random Forest (RF).

2.3.1. Support Vector Machines

SVM is a supervised learning algorithm that works by finding the optimal hyperplane that separates landslide-prone areas from non-prone areas in a high-dimensional feature space (Ageenko et al., 2022). SVM is particularly effective in handling non-linear relationships through the use of kernel functions, such as radial basis functions (RBF) or polynomial kernels. Its ability to perform well with small to medium-sized datasets makes it a popular choice for landslide susceptibility mapping (Huang et al, 2018). However, SVM can be computationally expensive, especially with large datasets, and requires careful tuning of hyperparameters to achieve optimal performance.

2.3.2. Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes (for classification) or the mean prediction (for regression) of the individual trees (Ageenko et al., 2022). RF is robust to overfitting and can handle large datasets with many variables, making it well-suited for landslide susceptibility mapping (Arabameri et al., 2020). One of the key advantages of RF is its ability to provide a measure of feature importance, which helps identify the most significant factors contributing to landslide risk (Ageenko et al., 2022). However, RF models can be less interpretable than simpler models like logistic regression, and their computational cost increases with the number of trees and features.

3. Materials and methods

3.1 Study Area

The Lagam Escarpment, part of the Elgeyo-Marakwet Escarpment (Kerio Escarpment), stretches across Keiyo, Marakwet, and West Pokot counties in Kenya. It forms the western boundary of the Kerio Valley depression, characterized by steep, rugged terrain with V-shaped valleys shaped by parallel rivers and streams. The escarpment reaches elevations of 3,520m ASL at Chemnirot and 3,500m ASL at Kamalogon Hill.

Geologically, the escarpment is composed of Basement System gneiss with porphyritic phonolite capping, while lateritic soils dominate except for alluvial deposits in valley floors. Sparse vegetation, aggravated by deforestation and agriculture, has left large sections bare and susceptible to landslides. Kaben Location is particularly landslide-prone, with major incidents such as the April 2020 disaster, where 26 people lost their lives due to landslides and flash floods (Jepkemboi et al., 2019).

This study will focus on the Lagam Escarpment area with the following spatial extent: Longitude: 35.45° to 35.6507°; Latitude: 0.98° to 1.2584°; Total Area: 692.64 km². It will cover parts of Endo, Embobut, Kapyego, Sambirir, Arror, Kapsowar and Moiben-Kuserwo Wards. Given the frequent landslides, ongoing terrain instability, and human-induced changes, the Lagam Escarpment presents a critical case for landslide susceptibility assessment and disaster risk management.

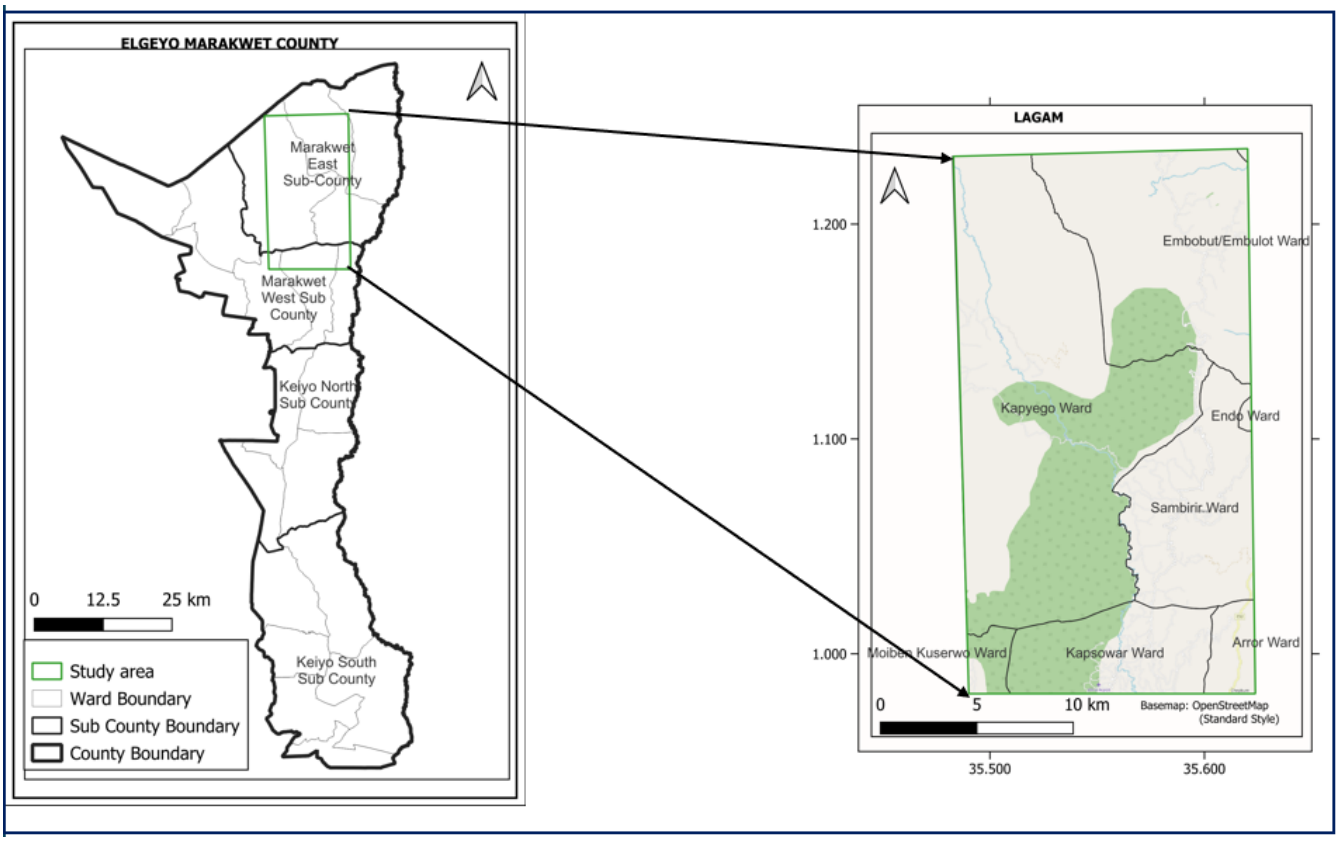


Figure 2: Map of study area

3.2. DATA

The data sets used in this study are displayed in Table 1 below.

Data Type	Source
Sentinel 1 SLC Interferograms (80m, 12days)	Alaska Site Facility (ASF) Vertex
Sentinel 2 (10m, 5 days)	Google Earth Engine
SRTM	USGS
CHIRPS (5000m, daily)	Google Earth Engine
Soil Data	ISRIC
Lithological and Geological Data	ISRIC
Roads	Humanitarian Data Exchange
Rivers	Humanitarian Data Exchange
Landslide Inventory	DRSRS and Elgeyo Marakwet County (GIS Department)

Table 1: Data Types and their Sources

The tools shown in Table 2 were instrumental for this study

Tool/Material	Role	Availability
Google Earth Engine (GEE)	Download and Processing	Freely Available
QGIS	Further Analysis & Maps	Free
Python (Jupyter Notebook)	Modelling and Analysis	Free

Table 2: Tools and Materials used in the study

3.3. METHODOLOGY

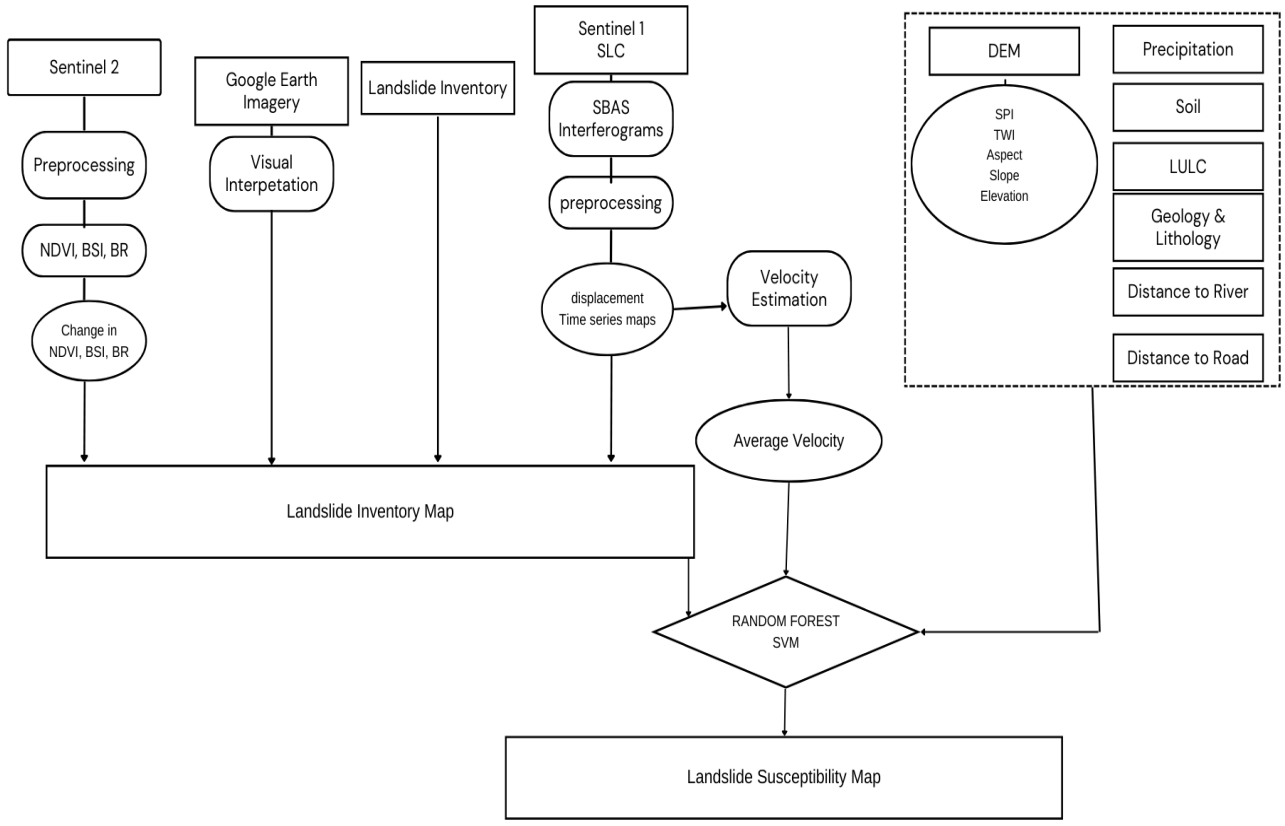


Figure 3: Overall Methodology Workflow

3.3.1. Landslide Mapping

3.3.1.1 Optical Remote Sensing Data (Sentinel-2)

The methodology for landslide detection and refinement is structured around multi-temporal analysis using Sentinel-2 data. The process consists of four key stages: pre- and post-event image processing, change detection, landslide extraction, and refinement using historical imagery.

Pre- and Post-Event Image Processing

Sentinel-2 surface reflectance data (COPERNICUS/S2_SR) were utilized for landslide detection. For each event year, cloud-masked pre- and post-event images were processed to generate median composites. The images were filtered using the Scene Classification Layer to remove clouds and shadows, ensuring high-quality observations. To enhance the detection of landscape disturbances, spectral indices were computed, including:

1. **Normalized Difference Vegetation Index (NDVI)**

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

High NDVI values indicate dense vegetation. A significant drop in NDVI suggests vegetation loss, which can be associated with landslides.

2. **Bare Soil Index (BSI)**

$$BSI = \frac{(SWIR + RED) - (NIR + BLUE)}{(SWIR + RED) + (NIR + BLUE)}$$

Higher BSI values indicate increased bare soil exposure. Landslides often leave behind bare soil, causing an increase in BSI.

3. **Brightness Ratio (BR)**

$$BR = \frac{RED}{RED + GREEN + BLUE}$$

Higher BR values suggest increased soil brightness, which can be indicative of newly exposed surfaces after a landslide.

4. **Normalized Difference Water Index (NDWI)**

$$NDWI = \frac{NIR - GREEN}{NIR + GREEN}$$

NDWI helps distinguish between water and land surfaces. A decrease in NDWI might indicate reduced soil moisture due to a landslide.

Change Detection and Landslide Indicator Mapping

Change indices were computed by differencing pre- and post-event values of the spectral indices. This allowed for the identification of areas with significant landscape changes, likely associated with landslide activity. Specifically, NDVI decline and BSI/BR increase were key indicators of landslide occurrence. The absolute differences in these indices were used to map the extent of possible landslide-affected regions.

NDVI Change: Large negative values indicate vegetation loss. (Figure 6)
BSI Change: Positive values indicate soil exposure. (Figure 5)
BR Change: Increased values indicate newly exposed bright soil. (Figure 4)
NDWI Change: Decreased values indicate drying of the surface. (Figure 7)

Landslide Extraction

Landslide-affected areas were delineated by applying threshold-based classification to the computed change indices. A connected component analysis was used to remove noise and isolate significant landslide patches. The threshold values were determined based on observed spectral differences between stable and disturbed areas.

NDVI Change < -0.45 → Indicates significant vegetation loss.
BSI Change > 0.1 → Indicates increased bare soil exposure.
BR Change > 0.03 → Indicates increased surface brightness.
NDWI Change < -0.03 → Indicate reduced moisture.

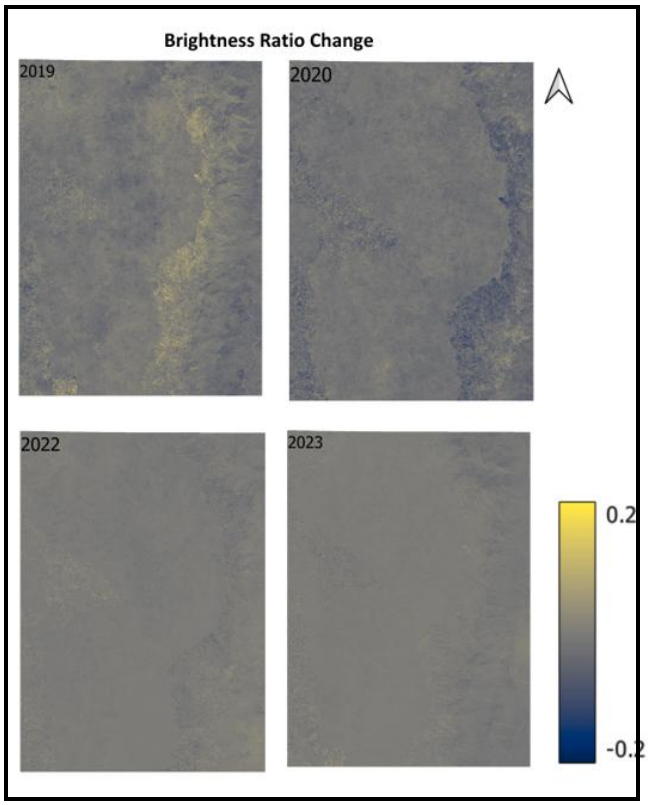


Figure 4: Change in Brightness Ratio

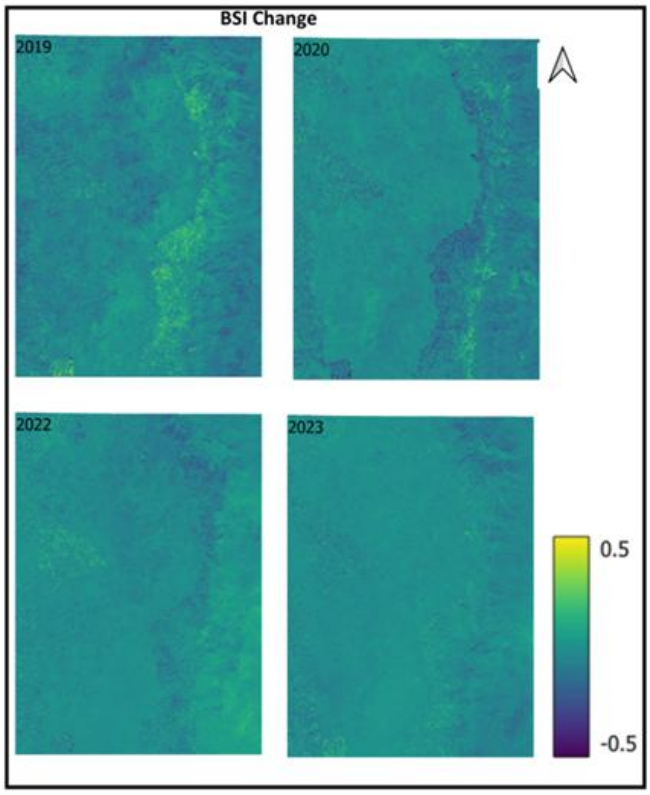


Figure 5: Change in Bare Soil Index

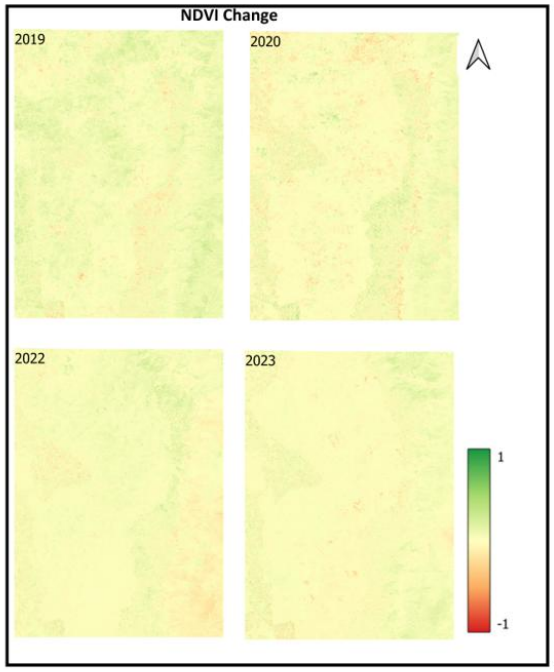


Figure 6: Change in Normalized Difference Vegetation Index

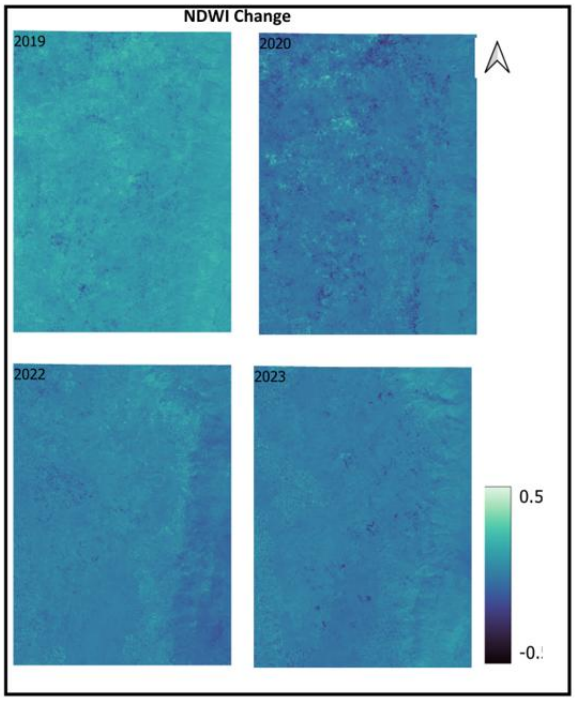


Figure 7:Change in Normalized Difference Vegetation Index

Refinement Using Historical Imagery

To improve the accuracy of landslide detection, refinement was performed using imagery from the year preceding each event. This step ensured that persistent land cover changes (e.g., deforestation or agricultural activities) were not misclassified as landslides. The refinement process involved:

- Temporal Consistency Check: NDVI trends from the preceding year were analyzed to differentiate landslides from seasonal vegetation changes.
- Texture Analysis: The Gray Level Co-occurrence Matrix (GLCM) contrast of the NIR band was computed to distinguish landslides from other land cover disturbances.
- Terrain Filtering: Elevation and slope constraints were applied to ensure detected areas were consistent with typical landslide-prone terrain.

Final Output

The refined landslide maps were generated for each event year, providing a comprehensive spatial and temporal analysis of landslide occurrences. In addition, change indices were exported for further statistical analysis and validation.

3.3.1.2 Google Earth Imagery Interpretation

High-resolution Google Earth imagery is visually interpreted to identify landslide occurrences. Landslide inventory is prepared based on the interpreted imagery. The landslide areas derived from Sentinel 2 were checked against the Google Earth Imagery for confirmation

3.3.1.3 SAR Data (Sentinel-1 SLC)

Sentinel-1 Single Look Complex (SLC) data was processed using the Small Baseline Subset (SBAS) interferometry approach to generate displacement time series maps.

A total of 602 interferograms were obtained from the Alaska Satellite Facility (ASF) using a baseline search. The study area is covered by flight path 130, with a temporal baseline of 36 days and a perpendicular baseline of 200m. The data spans from 2017-01-07 to 2024-07-17. The network is as shown in Figure 8 and Figure 9 below.

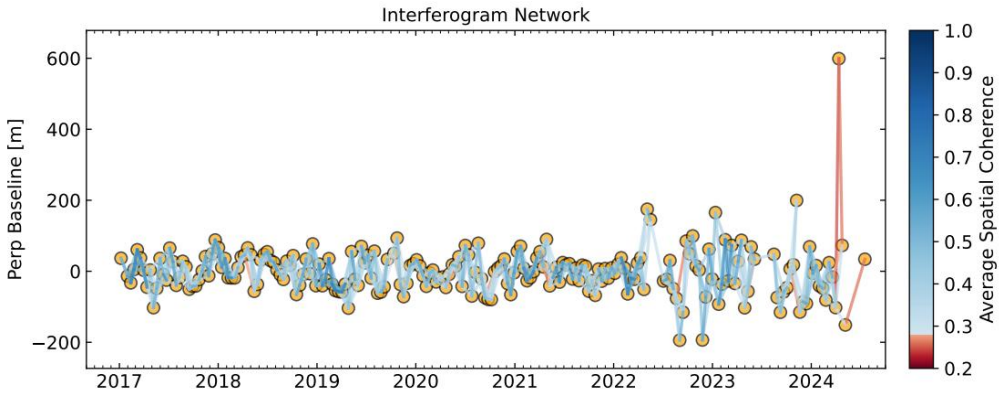


Figure 8: Interferogram Network

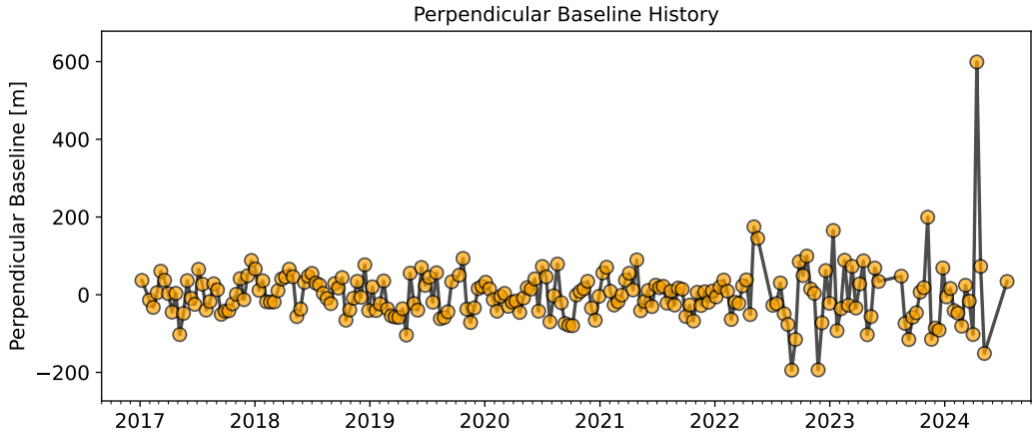


Figure 9: Perpendicular Baseline History

The area had an average coherence as depicted in the following (Figure 10 and Figure 11):

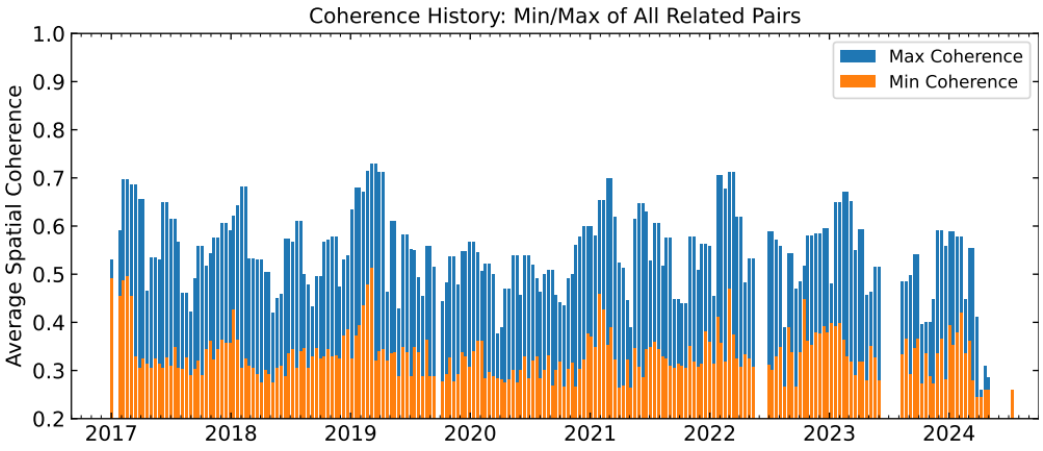


Figure 10: Coherence History

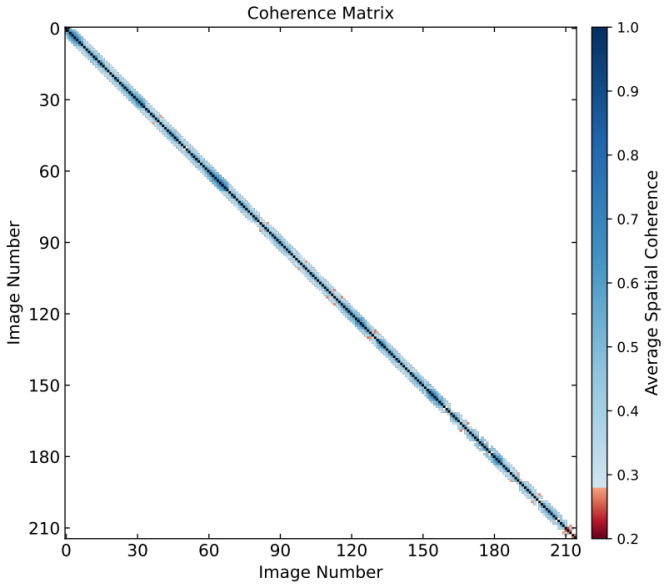


Figure 11: Coherence Matrix

The processing workflow follows a two-step correction approach as shown in Figure 12:

1. Phase Unwrapping and Inversion (Blue Ovals) – This step corrects unwrapping errors and inverts the raw phase time-series.
2. Noise Correction and Displacement Estimation (Green Ovals) – Corrections are applied to mitigate noise from different sources, leading to the final displacement time-series.

Certain steps in the workflow are optional, indicated by dashed boundaries.

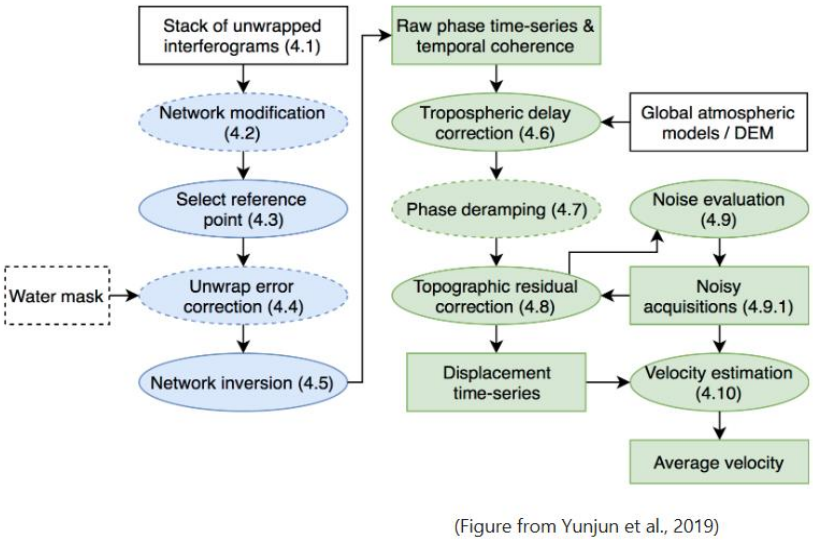


Figure 12: SBAS-InSAR Workflow

3.3.1.4. Landslide Inventory Mapping

- The results from optical (Sentinel-2) and SAR (Sentinel-1) analysis, along with visually interpreted landslide locations, are combined to create a comprehensive landslide inventory map.

3.3.2. Preparation of Conditioning Factors

- Various conditioning factors influencing landslide susceptibility are collected and prepared:
    - **Topographical Factors (Derived from DEM(Figure 13)):**
        - Stream Power Index (SPI)
        - Topographic Wetness Index (TWI)
        - Aspect: Eastness and Northerness
        - Elevation

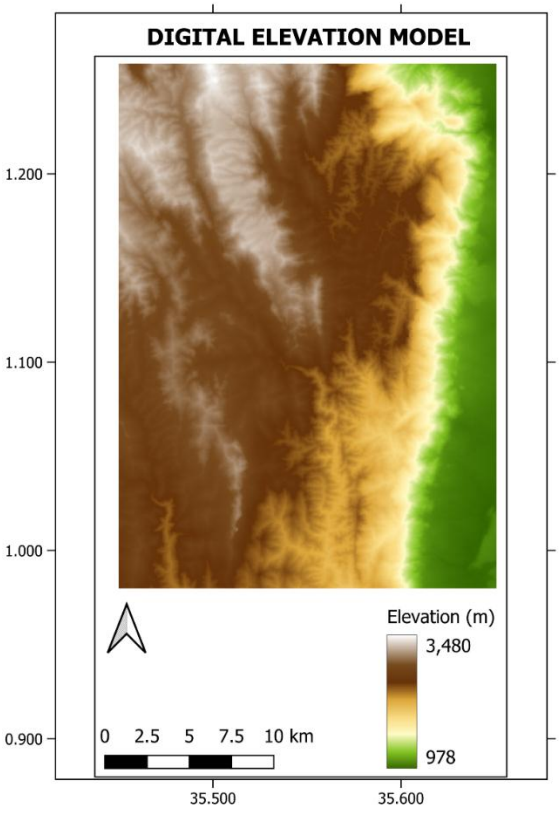


Figure 13: Digital Elevation Model

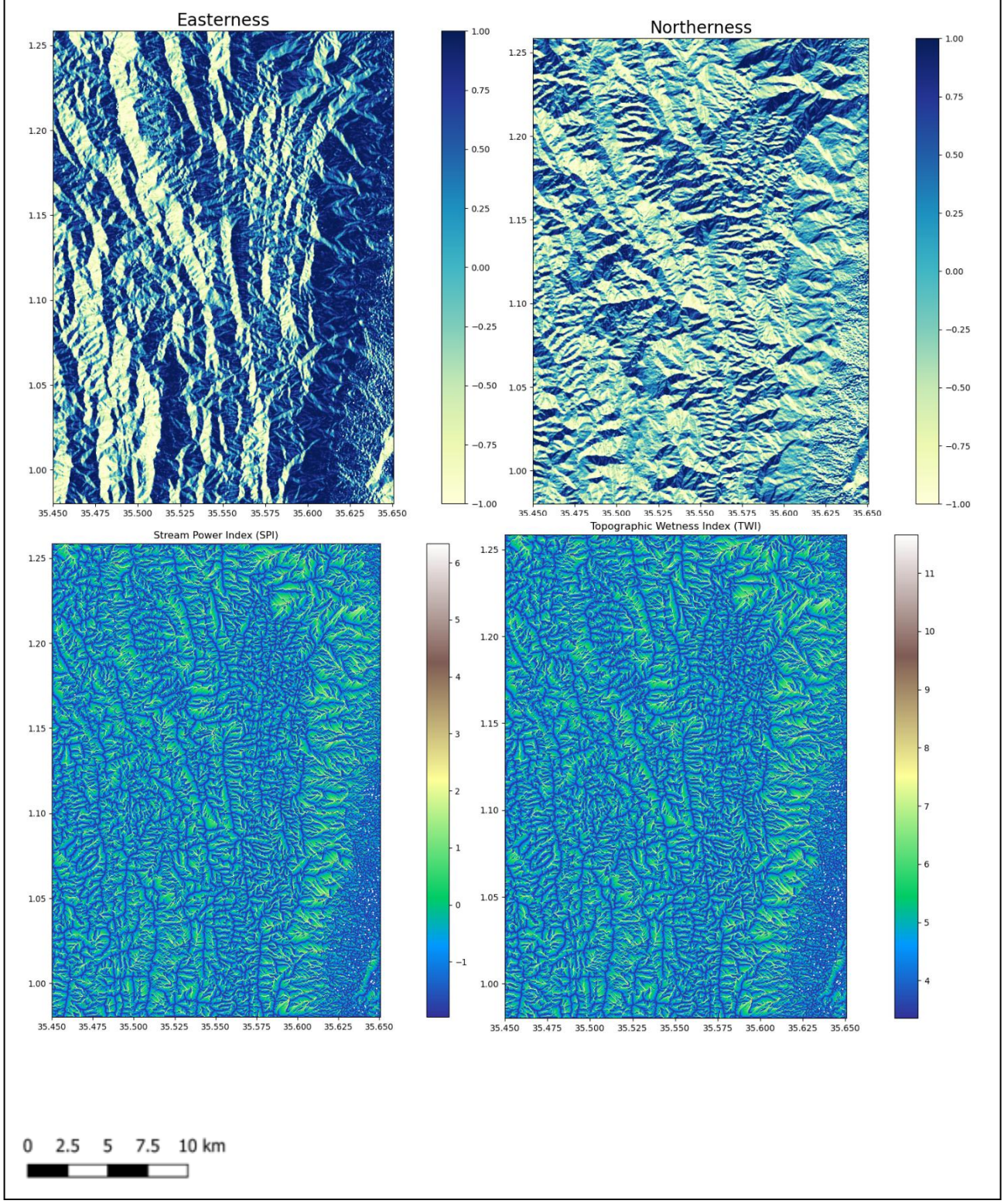


Figure 14: Easternness, Northerness, Topographic Wetness Index (TWI) and Stream Power Index (SPI)

- **Climatic Factors:** Precipitation

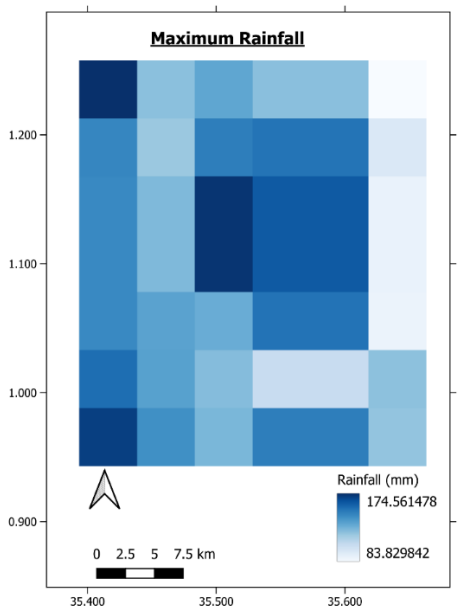


Figure 15: Maximum Rainfall

- **Soil Characteristics**

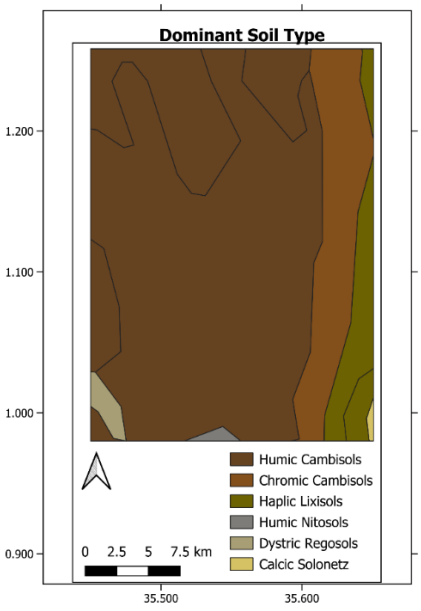


Figure 16: Soil Type

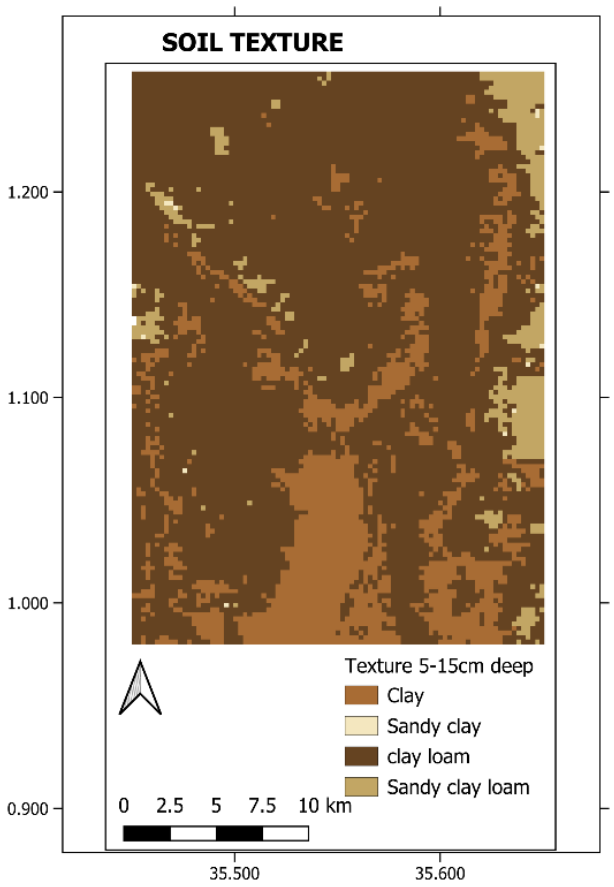


Figure 17: Soil Texture 5-15 cm Deep

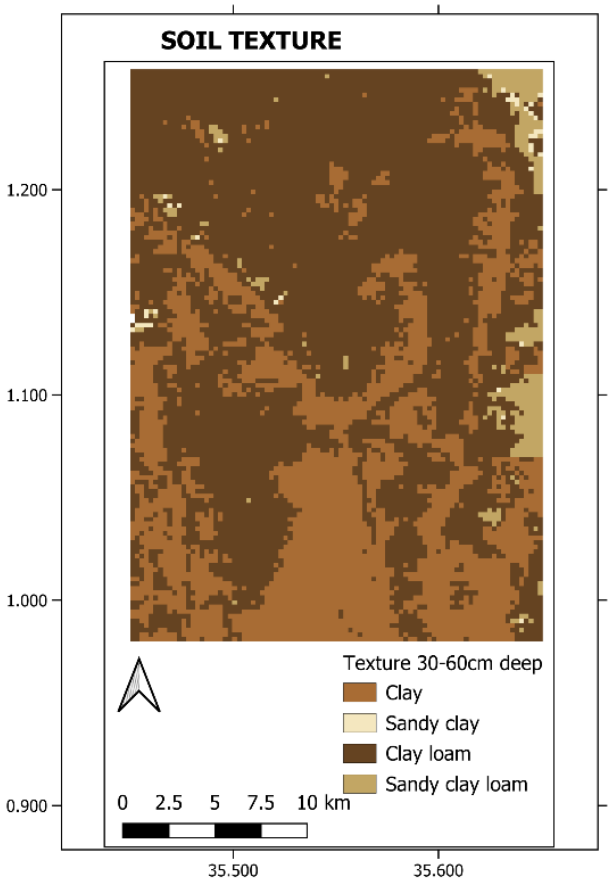


Figure 18: Soil Texture 30-60 cm Deep

- **Land Use/Land Cover (LULC)**

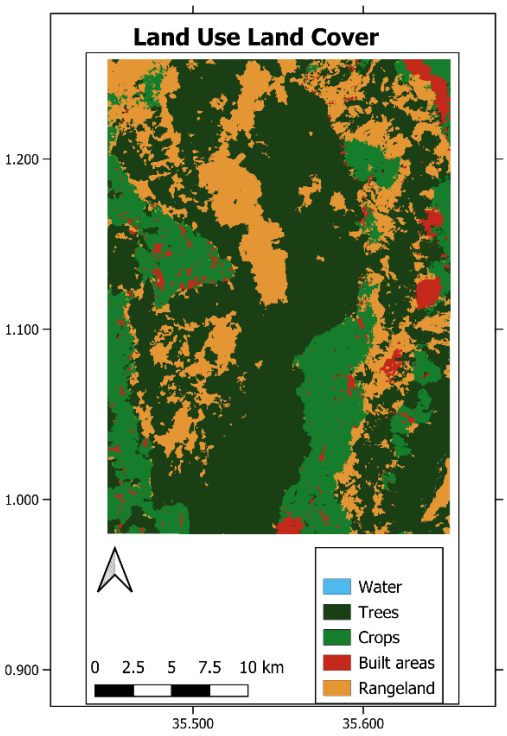


Figure 20: Land Use Land Cover Map

Geology and Lithology

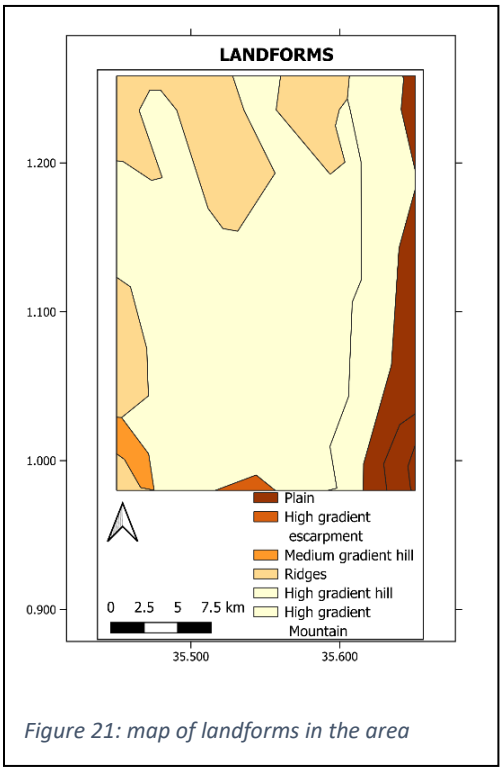


Figure 21: map of landforms in the area

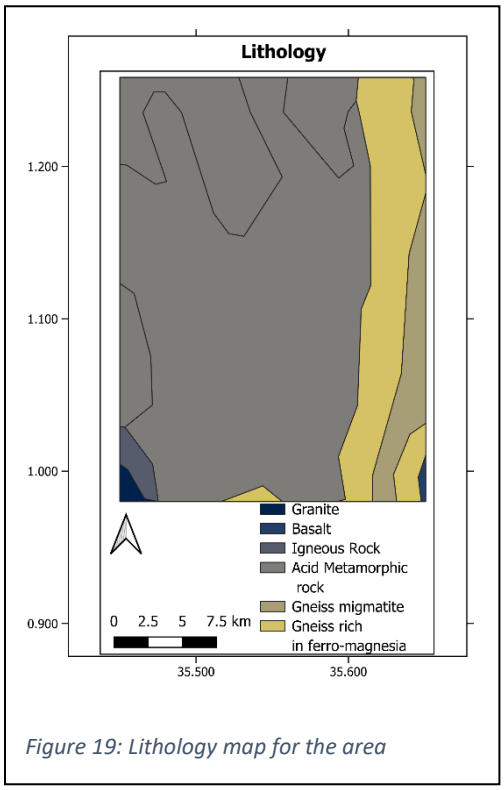


Figure 19: Lithology map for the area

- **Proximity Factors:** Distance to rivers and roads

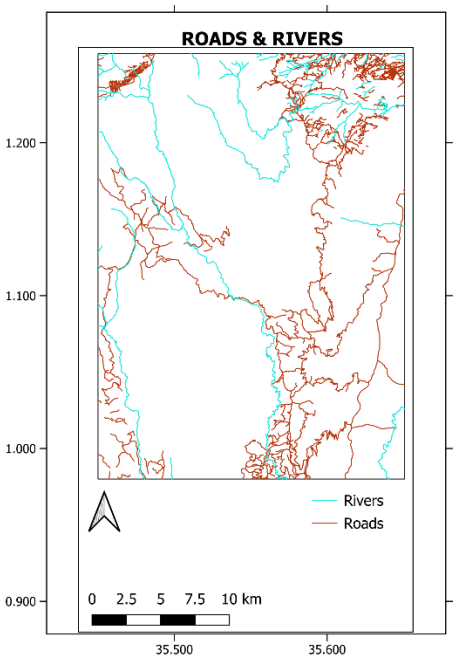


Figure 22: Roads and Waterways present in the area

- **Historical landslides and field data**

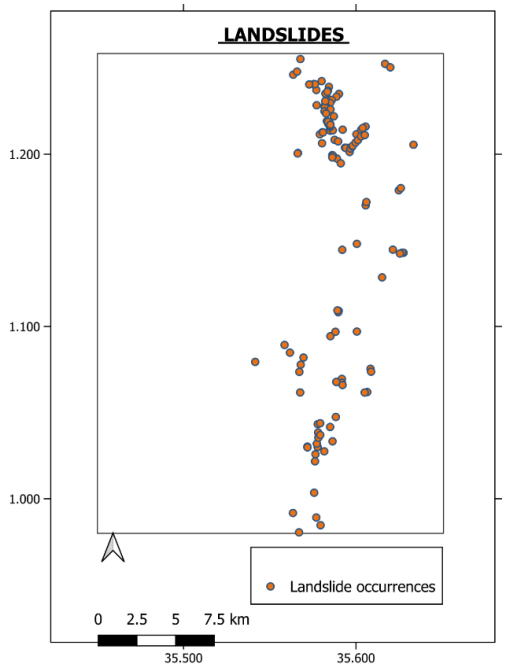


Figure 23: Recorded Landslide Occurrences in Lagam

3.3.3. Landslide Susceptibility Modeling

Two machine learning algorithms were employed for landslide susceptibility modeling: Random Forest (RF) and Support Vector Machine (SVM). The RF algorithm operates as an ensemble learning method that constructs multiple decision trees and aggregates their predictions, utilizing bootstrapping (random sampling with replacement) and feature randomness to enhance generalization and reduce overfitting. This approach is particularly advantageous for landslide modeling as it effectively handles non-linear relationships between conditioning factors, provides quantifiable feature importance scores (e.g., evaluating the relative impact of rainfall versus slope angle), and demonstrates robustness against noisy input data. The SVM algorithm functions as a supervised learning model that identifies an optimal hyperplane to classify areas into susceptible and non-susceptible categories, employing kernel functions such as the radial basis function to manage complex, non-linear decision boundaries. SVM offers particular strengths in high-dimensional feature spaces (accommodating numerous conditioning factors) and performs reliably with small to medium-sized datasets.

During model development, the labeled landslide inventory data (historical landslide occurrences) was used to train both algorithms to recognize predictive patterns among the various conditioning factors. Model performance was rigorously evaluated using multiple validation metrics, including classification accuracy (percentage of correctly classified pixels), precision-recall balance (addressing false positives and negatives), and the Area Under the Receiver Operating Characteristic curve (AUC-ROC), where values approaching 1.0 indicate superior predictive capability.

3.3.4. Generation of Landslide Susceptibility Map

The trained machine learning models (RF and SVM) predict landslide occurrence probabilities for each pixel across the study area. These probabilities are classified into five susceptibility zones depending on percentage. The resulting predictions are exported to GIS software QGIS to generate a raster-based susceptibility map, where risk levels are visually distinguished using a color gradient. This final susceptibility map serves as a critical tool for identifying hazard-prone areas and supporting mitigation strategies.

4. RESULTS AND DISCUSSION

4.1. DISPLACEMENT MAPS

The displacement in the region varied from season to season, with positive displacement being witnessed during rainy season as vegetation grew and negative displacement during dry season when there is a few vegetation and some of the vegetation being cleared as land is prepared for cultivation. The figure below indicates that the negative values are moving away from the satellite while the positive values are moving towards the satellite. Displacement ranged from -30 cm to +30 cm as displayed in Figure 24 and Figure 25.

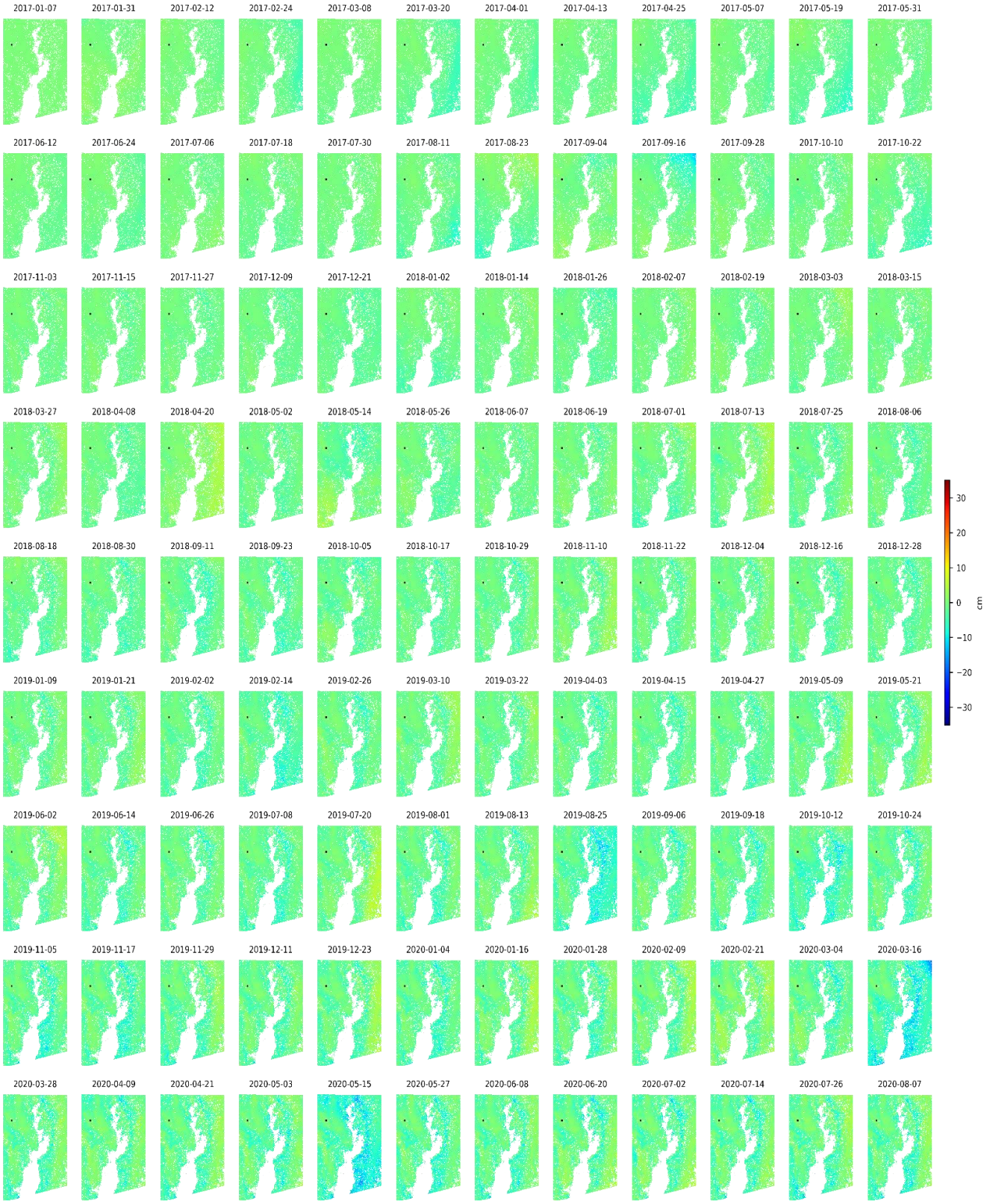


Figure 24: Displacement Maps for 2017- 2020

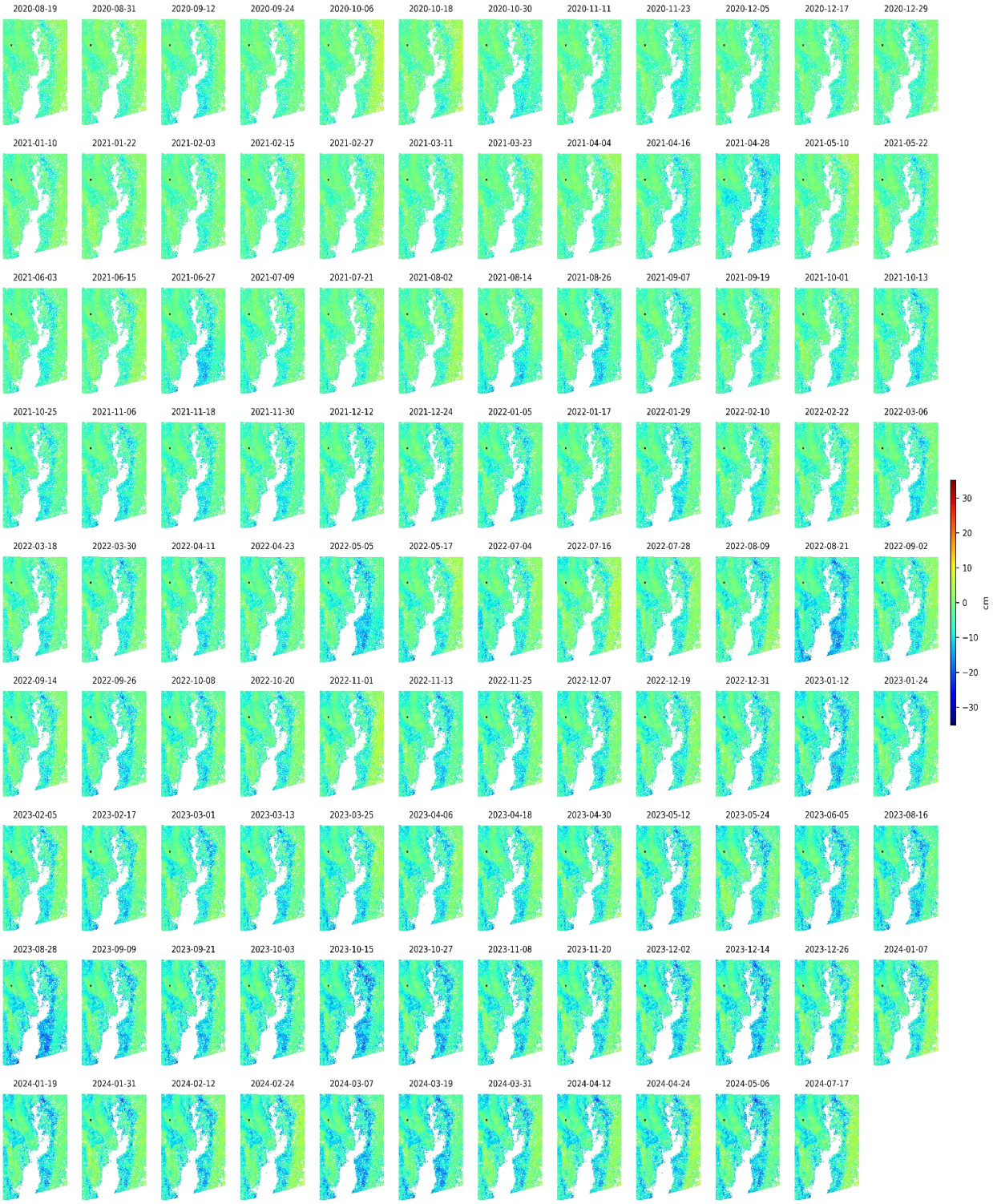


Figure 25: Displacement Maps for 2020 - 2024

4.2. VELOCITY MAP

The deformation rate of the area is -4cm/year to 3cm/year as shown in Figure 26 below. Since the area is hilly with steep terrain as it is an escarpment, and part of it is covered by forest, and also has a high vegetation cover, it suffers from decorrelation. Some regions with actual deformation may be masked and obscured due to the coherence mask processing as the threshold was set to 0.4.

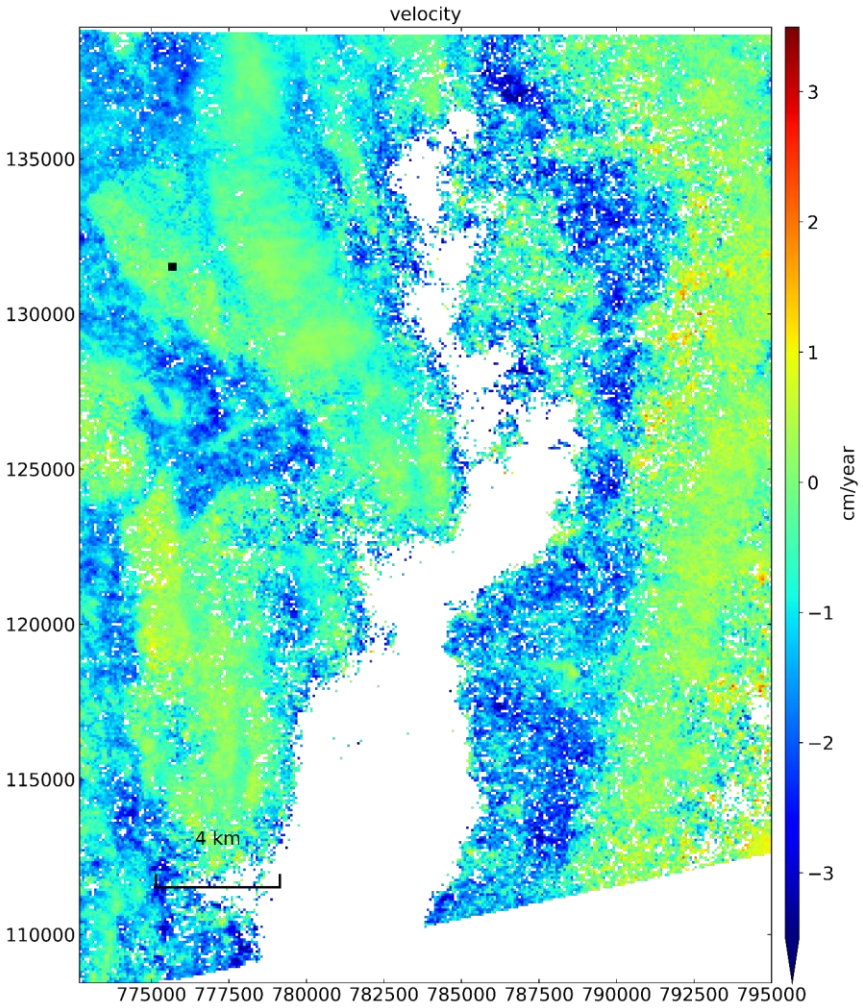


Figure 26: Velocity map for Lagam from 2017 - 2024

4.3. LANDSLIDE MAPS

The landslide maps were created using both optical and InSAR data. Figure 27 below indicates the displacement over time for one of the areas that was affected landslide. From the figure, we can see that after the landslide occurrence in 2020, the place has been continually deforming.

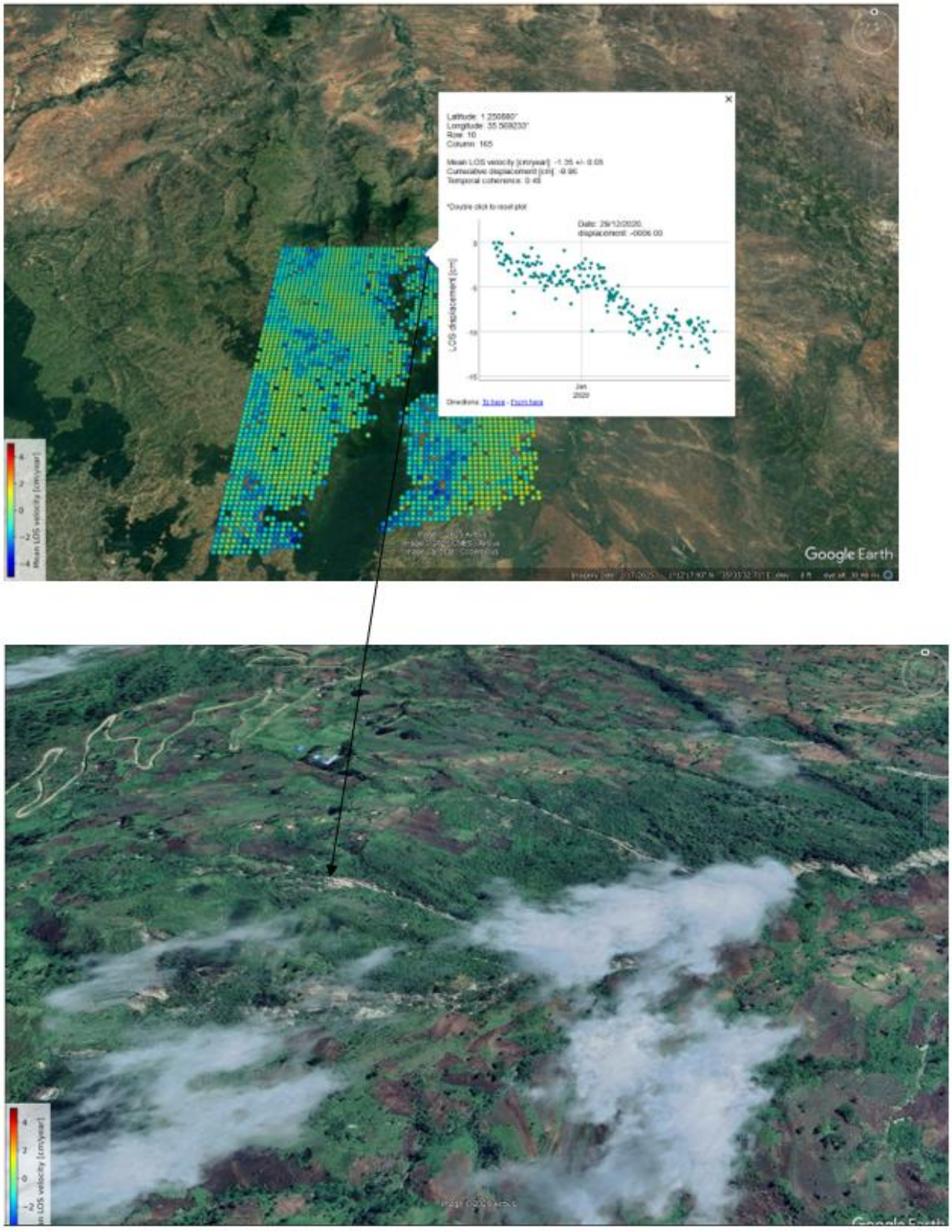


Figure 27: Velocity map displayed in Google Earth Pro and the Landslide image of the same area

4.3.1. Updated landslide inventory map

Most landslides displayed in Figure 28 were mapped from the optical and radar data were those that occurred along the escarpment. This is in line with what had been collected in the field by the County's GIS Department. A few landslides were detected in other areas, and their deformation rates and google earth imagery were used to verify their occurrence.

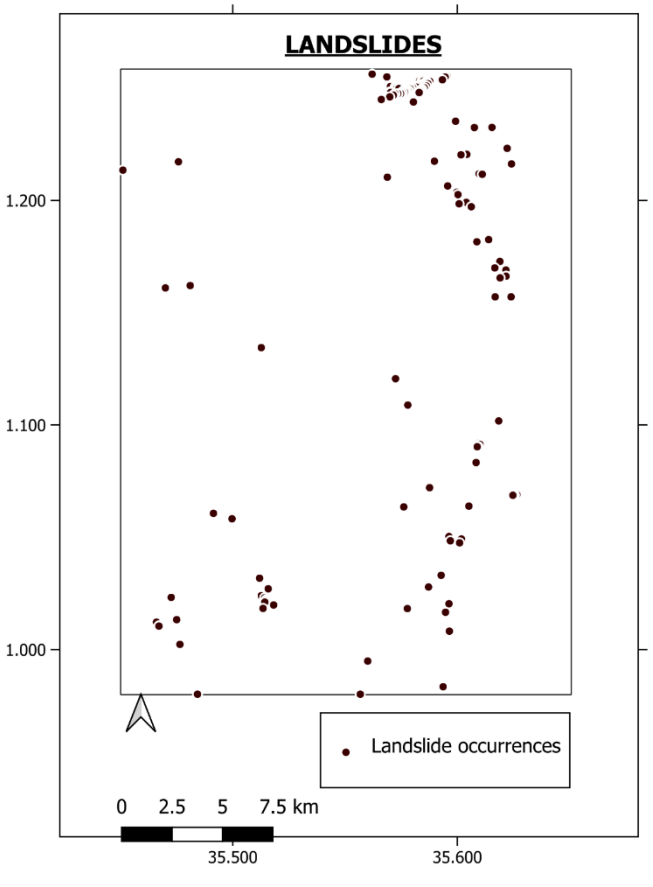


Figure 28: New Landslide inventory Map

4.4. LANDLIDE SUSCEPTIBILITY MAPS

The landslide susceptibility maps were generated using two different machine learning models: Random Forest (RF) and Support Vector Machine (SVM). These maps visualize the spatial distribution of landslide susceptibility across the study area, categorizing regions into different risk levels based on the probability of landslide occurrence.

4.4.1. Random Forest

The Random Forest's Landslide Susceptibility Map in Figure 29 divides the study area into five susceptibility classes: very high (90-100%), high (80-90%), moderate (60-80%), low (40-60%), and very low (0-40%). The map highlights areas with the highest susceptibility (90-100%) as the most vulnerable to landslides, which are critical for prioritizing mitigation measures. The scale of the map (0 to 7.5 km) provides a clear understanding of the spatial extent of the analysis. The Random Forest model's ability to handle complex, non-linear relationships in the data is reflected in the detailed susceptibility classification.

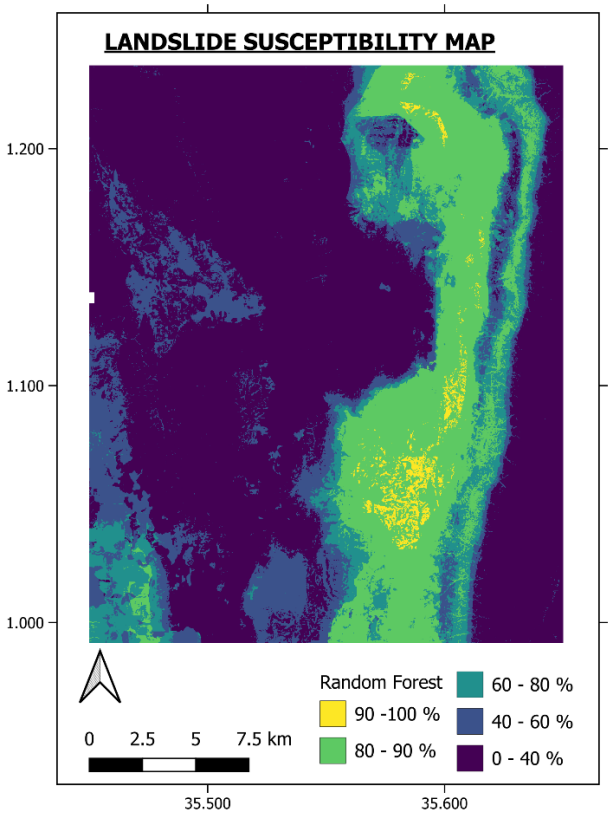


Figure 29: Random Forest's Landslide Susceptibility Map

4.4.2. SVM

The SVM Landslide Susceptibility Map in Figure 30 also categorizes the study area into five susceptibility classes: very high (90-100%), high (80-90%), moderate (60-80%), low (40-60%), and very low (0-40%). Similar to the Random Forest map, areas with very high susceptibility (90-100%) are identified as the most at risk. The SVM model, known for its effectiveness in high-dimensional spaces, provides a slightly different spatial distribution of susceptibility compared to the Random Forest model. This variation highlights the importance of using multiple models to capture different aspects of landslide susceptibility

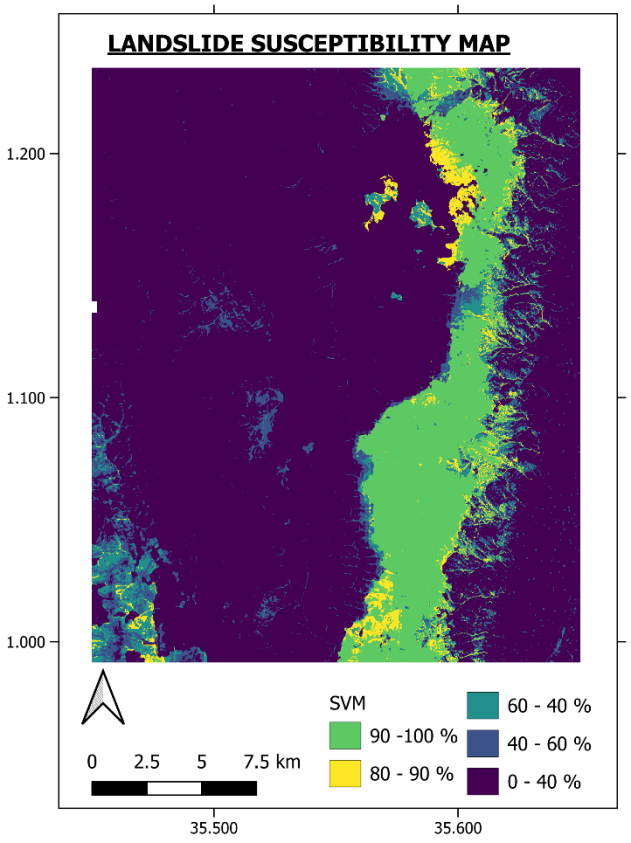


Figure 30: SVM's Landslide Susceptibility Map

Comparing the two maps, both the Random Forest and SVM models identify similar regions with very high susceptibility (90-100%), indicating consistency in predicting the most vulnerable areas. However, there are notable differences in the spatial distribution of moderate and low susceptibility classes.

The Random Forest model tends to produce more granular predictions due to its ensemble nature, while the SVM model provides a smoother distribution of susceptibility classes, reflecting its ability to handle complex decision boundaries. These differences underscore the complementary nature of the two models and the value of using multiple approaches for landslide susceptibility mapping.

4.5. ACCURACY ASSESSMENTS FOR LANDSLIDE SUSCEPTIBILITY MAPS

The performance of each model was evaluated and they are demonstrated below. Random Forest showed better performance than SVM.

4.5.1. Random Forest

The performance of the Random Forest (RF) classifier in predicting landslide susceptibility was evaluated using a confusion matrix as shown in Figure 31. The matrix revealed that the model correctly predicted 42 instances of non-landslide areas (True Negatives) and 7 instances of landslide areas (True Positives). However, there were 31 instances where the model failed to identify actual landslide areas (False Negatives), and 5 instances where non-landslide areas were incorrectly classified as landslide areas (False Positives). These results indicate that while the model has a good ability to identify non-landslide areas, there is room for improvement in accurately detecting landslide-prone areas. The presence of false negatives is particularly critical in landslide susceptibility mapping, as it represents areas at risk that the model failed to identify.

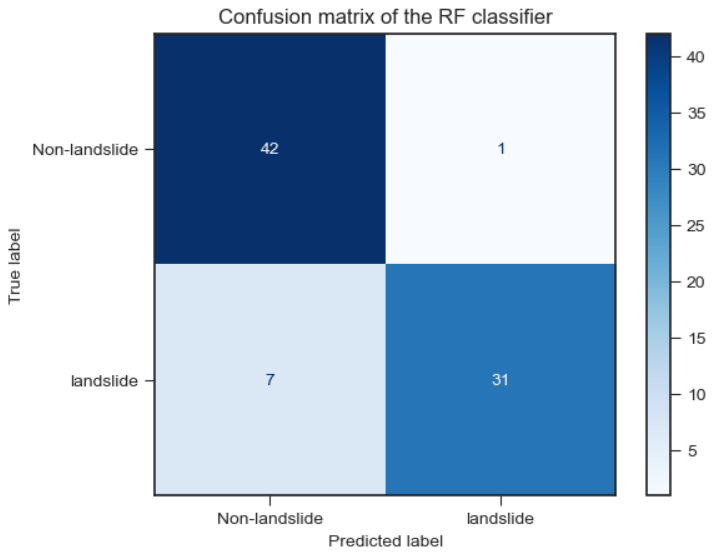


Figure 31: Confusion matrix of the RF classifier

To optimize the hyperparameters of the model, a GridSearchCV approach was employed, which systematically explored various parameter combinations to enhance model performance. The model's effectiveness was assessed using the Area Under the Curve (AUC) metric from the Receiver Operating Characteristic (ROC) curve in Figure 32, achieving an AUC score of 0.97. This high AUC value indicates excellent discriminatory power, with the model effectively distinguishing between landslide and non-landslide areas. The True Positive Rate (TPR) and False Positive Rate (FPR) were also analyzed to understand the model's sensitivity and specificity. The high AUC score underscores the model's robustness and reliability in landslide susceptibility mapping, making it a valuable tool for risk assessment and mitigation planning.

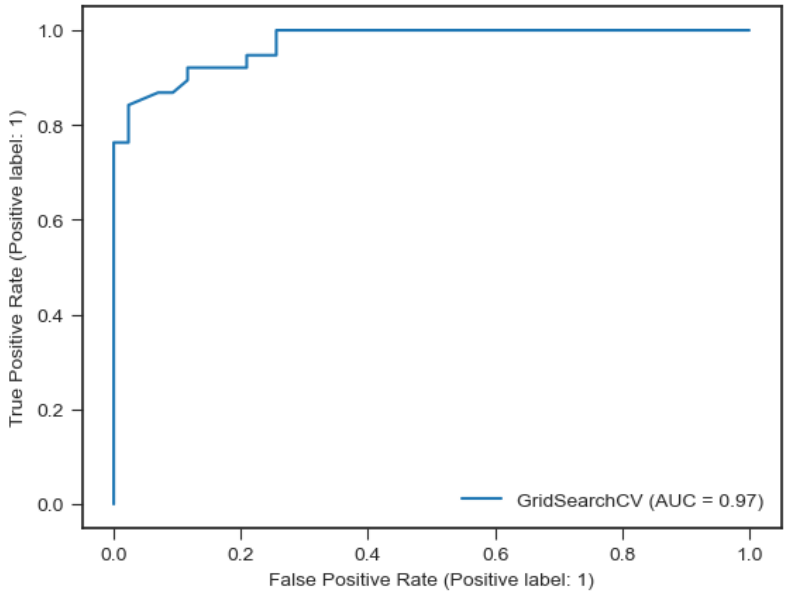


Figure 32: Random forest's ROC curve

4.5.2. Support Vector Machine (SVM)

The performance of the Support Vector Machine (SVM) classifier in predicting landslide susceptibility was evaluated using a confusion matrix in Figure 33. The matrix revealed that the model correctly predicted 42 instances of non-landslide areas (True Negatives) and 6 instances of landslide areas (True Positives). However, there were 32 instances where the model failed to identify actual landslide areas (False Negatives), and 5 instances where non-landslide areas were incorrectly classified as landslide areas (False Positives). These results indicate that while the model has a good ability to identify non-landslide areas, there is room for improvement in accurately detecting landslide-prone areas. The presence of false negatives is particularly critical in landslide susceptibility mapping, as it represents areas at risk that the model failed to identify

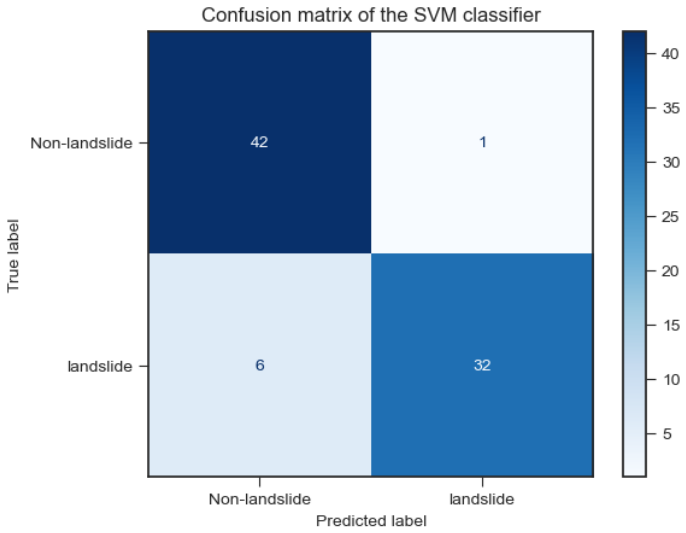


Figure 33: Confusion matrix of the SVM classifier

The performance of the model was further evaluated using the True Positive Rate (TPR) and False Positive Rate (FPR), with an Area Under the Curve (AUC) score of 0.95 obtained through a GridSearchCV process as shown in Figure 34. The high AUC score indicates that the model has excellent discriminatory power, effectively distinguishing between landslide-prone and non-landslide areas. The TPR, which measures the proportion of actual landslide areas correctly identified by the model, was analyzed alongside the FPR, which measures the proportion of non-landslide areas incorrectly classified as landslide-prone. The AUC score of 0.95 underscores the model's robustness and reliability in landslide susceptibility mapping, making it a valuable tool for risk assessment and mitigation planning.

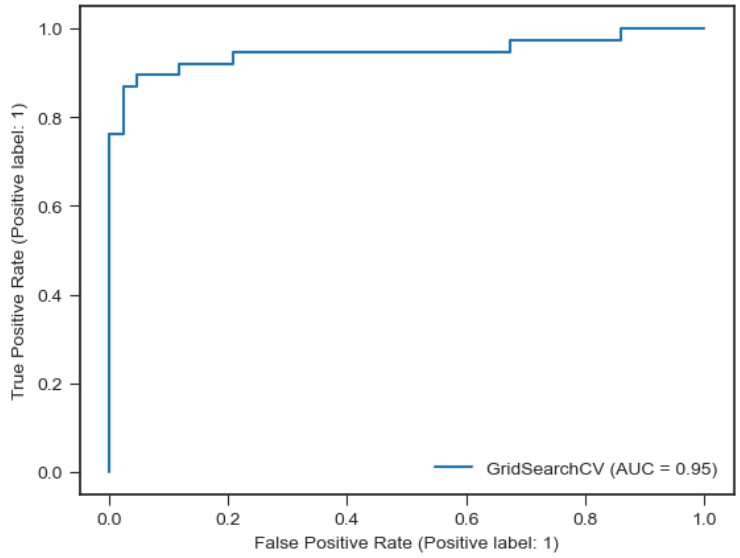


Figure 34: SVM's ROC curve

5. CONCLUSION AND OUTLOOK

In this study, I explored the effectiveness of SBAS-InSAR techniques applied to Sentinel-1 data in detecting and monitoring ground displacement in the Lagam Escarpment. My results demonstrated that SBAS-InSAR effectively captured surface deformation trends over time, highlighting regions with significant displacement that may indicate potential landslide activity. The observed patterns provided valuable insights into the temporal evolution of ground movement, reinforcing the capability of InSAR for long-term monitoring in landslide-prone regions.

Additionally, I integrated SBAS-InSAR with Sentinel-2 optical imagery to enhance the identification and mapping of past landslide occurrences. By combining ground deformation data with surface reflectance changes, I improved the accuracy of landslide inventory mapping, particularly in areas where optical imagery alone was insufficient due to vegetation cover or cloud interference. This multi-sensor approach proved crucial in distinguishing landslides from other geomorphic changes in the study area.

Furthermore, I demonstrated that integrating SBAS-InSAR results, optical imagery-derived landslide inventories, and machine learning models—Support Vector Machine (SVM) and Random Forest (RF)—significantly improved landslide susceptibility assessments. By combining geospatial datasets with advanced classification techniques, I developed a more reliable and data-driven approach for identifying high-risk zones. My hybrid framework effectively captured complex relationships between landslide conditioning factors, leading to refined susceptibility maps that can support risk mitigation and land-use planning in the Lagam Escarpment.

Overall, my research underscores the value of remote sensing and machine learning in landslide detection and susceptibility assessment. The findings contribute to a more comprehensive understanding of landslide dynamics and offer practical insights for disaster risk management in similarly geologically sensitive regions.

Recommendations

To further enhance the accuracy and applicability of landslide hazard assessment, I propose the following recommendations. First, the use of high-resolution Digital Elevation Models and optical data should be prioritized. High-resolution DEM, such as those derived from LiDAR or drone imagery, can provide more detailed topographic information, improving the identification of landslide-prone areas. Similarly, high-resolution optical data can capture finer details of surface changes, enabling more precise mapping of landslide susceptibility.

Second, the exploration of drone imagery in landslide mapping should be pursued. Drones offer the ability to collect high-resolution, real-time data in areas that are difficult to access, while LiDAR provides detailed 3D terrain models that can reveal subtle ground deformations. These technologies can complement SBAS-InSAR and optical imagery, providing a more comprehensive understanding of landslide dynamics.

Third, the use of other SAR data sources, such as ALOS PALSAR, should be explored. ALOS PALSAR offers longer wavelength SAR data, which can penetrate dense vegetation and provide additional insights into ground deformation. Integrating multiple SAR datasets can enhance the robustness of landslide monitoring and improve the accuracy of susceptibility maps.

Finally, the application of deep learning models should be investigated. Deep learning techniques, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have shown great promise in handling complex, high-dimensional datasets. These models can potentially improve the accuracy of landslide predictions by capturing more intricate patterns and relationships in the data.

In conclusion, this study has advanced the field of landslide hazard assessment by demonstrating the potential of integrating SBAS-InSAR, optical imagery, and machine learning. The results provide a robust framework for landslide risk management, contributing to the safety and resilience of communities in the Lagam Escarpment and beyond.

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